

UAD MODEL EXTENSION

This blender plugin provides easy access to creating models that can be used with the Ultimate Admiral Dreadnoughts Mod.

Functions

- Node creation, including: GameObject, place:flag, SmokePlaceholder, Collider, Mount, Decor, Variation, HullInfo, and Mesh
- Model export: export the selected model as Wavefront obj and model description json
- Model import: import the model through the model's description json.

Precautions

- When using, use the blender coordinate system (Y forward, Z up, right hand system), including the values of each internal attribute, and the coordinate system rotation will be automatically completed when importing and exporting.
- All model bows point forward (+Y).
- When exporting, select all the nodes that need to be exported, click File> export > UAD Models, and then create a new folder and save the json in that folder. Mods locate content by the name of that folder when looking for model names in parts.csv and partModels.csv.
- Only one model can be exported at a time.

Introduction to nodes

1. GameObject

- Ordinary nodes, corresponding to a GameObject in Unity
- In this plugin, it is shown as a pure empty object with a size of 0.5 m

2. place:flag

- The flag's placeholder node will be replaced with the flag in UAD
- The name must be place:flag (this plugin ignores everything after the ".").
- The node position corresponds to the flagpole side at the top of the flag, and the upward axis is parallel to the flagpole
- This plugin is shown as a cube of size 0.5m

3. SmokePlaceholder

- The smoke occupancy node of the funnel, from which the smoke is expelled during the battle
- The name must be SmokePlaceholder, (this plugin ignores everything after the ".").
- Theoretically, only position matters

- This plugin is shown as a cube of size 0.5m

4. Collider

- The Hitbox nodes correspond to each DeckSize, HangSize, DeckWall, DeckPlace, and DeckBorder in UAD
- The size attribute is fixed to (1, 1, 1), the center attribute is fixed to (0, 0, 0), and the size and position are adjusted using the node's transformation and scale
- In this plugin, it is displayed as a cube of size 0.5m (corresponding to the size of the BoxCollider in unity).

5. Mount

- Anchor nodes
- For the meaning of each attribute, please refer to mounts.csv from TAF
- This plugin is shown as a spherical empty object with a size of 0.5m

6. Decor

- The decorator node, where the contents under it are hidden (not rendered) when colliding with other parts
- In this plugin, it is shown as a pure shaft empty object with a size of 0.5 m
- The currently speculated collision inspection mechanism is as follows:
 - a. {hull} and forceOverlap are used to get the parts type that ignores collisions, ignoreTypes
 - b. All the meshes under the Decor node are found to be the overall AABB hitbox.
 - c. Scale the hitbox with maxOverlapSizeBox to detect if it collides with other parts (excluding part types in ignoreTypes);
 - d. If it is not detected, scale the hitbox at $0.5 \times \text{minOverlapSizeBox}$ to detect if it collides with other parts (excluding components in ignoreTypes, and must be a funnel, tower_main, or tower_sec).

7. Variation

- Variant nodes, available only for hulls, select one child node among multiple child nodes to render
- default: Default to choose which variant, if empty, the first
- random: Whether it is randomly selected
- variations: the names of the individual variants, separated by commas
- Each child node is recommended to be consistent with the content in Variations (this plugin ignores everything after the ".").
- This plugin is shown as a spherical empty object with a size of 0.5m

8. HullInfo

- The root node of the hull model
- waterLevel: function unknown. In the original model, akizuki_hull_b is -0.5, jap_tb_hull is -1.25, and the rest are 0
- stabilityForward: Used to adjust the position of the ship's center of gravity
- schemes: The function is unknown, and the non-empty values of this item in the hull model in the original model are far_sides and side_main_guns, corresponding to the model such as below:
 - side_main_guns: gazelle_hull_var1, gazelle_hull, emden_hull_var2, emden_hull_var, emden_hull, monitor_hull, virginia_hull, gazelle_hull_mast_b, gazelle_hull_mast_a, gazelle_hull_mast_d, gazelle_hull_tall, gazelle_hull_mast_c, gazelle_hull_mast_a_var
 - far_sides: tsesarevich_hull_a, borodino_hull_a

9. Mesh

- Model nodes
- Meshes are mapped in this plugin
- When exporting, each Mesh data is exported as a Wavefront obj file named after the data name (ignores content after the "."; if the names are the same, only one of them will be exported).
- The model should have multiple vertex groups, each of which is assigned a material, and currently only supports the component materials that come with UAD. builtin_materials.blend provides a simple version of these materials is provided for previewing UVs

Introduction to model hierarchy

Turret

The typical structure of the turret model is as follows:

- part
 - Visual
 - turret_model
 - barrel_model_1
 - ...
 - other_models
 - DeckSize
 - ...
 - HangSize
 - ...

- Decors & Mounts

1. DeckSize is the Collider node, which represents the turret-to-deck contact part, and the recommended height scaling is 0.1
2. HangSize is the Collider node, which represents the projection of the turret and the non-contact parts of the deck (such as gun barrels) on the deck, and the height scaling is recommended to be 0.1
3. The position of the root node of the model corresponds to the intersection of the axis of rotation and the deck (also the anchor position).
4. The origin of the gun model should be on the pitch axis
5. To ensure that the gun barrel is properly bound, the gun model name must contain barrel, and Visual suggests that all gun barrel parts are directly under the turret node

Funnel

The typical structure is as follows:

- part
 - Visual
 - models
 - SmokePlaceholder
 - DeckSize
 - ...
 - Decors & Mounts

1. DeckSize is the Collider node, which represents the funnel-deck contact part, and the height scaling recommendation is 0.1
2. The location of the root node of the model corresponds to the anchor position of the funnel

Towers

The typical structure is as follows:

- part
 - Visual
 - models
 - place:flag

- ...
- DeckSize
- ...
- Decors & Mounts & DeckWalls

1. DeckSize is the Collider node, which represents the bridge and deck contact part, and the height scaling recommendation is 0.1
2. The DeckWall is a Collider node, and components cannot pass through it, so it can be used to limit the rotation angle of turrets of different sizes in the anchor point
3. The location of the root node of the model corresponds to the anchor point position of the bridge where the bridge is placed

Hulls

The typical structure is as follows

- HullInfo
 - Visual
 - Sections
 - Bow
 - Middle1
 - ...
 - Stern

Bow, Middle1, Stern and other nodes each correspond to a section of the hull, and the structure is as follows:

- Bow
 - Variation
 - var_1
 - Meshes and Colliders and Mounts and place:flags
 - ...
 - Meshes
 - Mounts
 - Deck
 - DeckPlace
 - DeckBorderLeft
 - DeckBorderRight
 - DeckBorderFront
 - DeckBorderBack

- DeckFake
 - ...
- DeckWall

1. The Variation node is recommended to be named Variation
2. The DeckPlace node is a Collider node, which means the deck can be placed in the area, and the height scaling recommendation is small
3. The DeckBorderXXX node is a Collider node, which represents the deck boundary, referring to the red diagonal stripe part when the reason for the border cannot be placed
4. The DeckWall node is a Collider node, which is a "wall" set to prevent mold penetration during rotation and placement
5. Deck is a GameObject node, and it is recommended to place DeckPlace and DeckBorderXXX below it (both optional) when organizing the model
6. The structure of DeckFake is the same as that of Deck, and its specific function is still to be analyzed
7. The model position should be on the water surface and the central axis of the hull, and the position on the forward axis will not affect the final effect